Supplementary Material

to the article

"Multiplying Bayes Factors Across Studies: Which Hypotheses Are We Testing?"

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This supplement consists of two parts. Section S1 derives expressions for $\mathrm{BF}^{(\text{joint})}_{c^*,u}$, the ratio of $\mathrm{BF}^{(\text{joint})}_{1,u}$ to $\mathrm{BF}^{(\text{joint})}_{1,c^*}$, and uses them to examine whether the two Bayes factors point in the same direction and under which conditions their ratio approaches one as studies are added (main paper, Section 7.2). Section S2 extends the computation of $\mathrm{BF}^{(\text{joint})}_{1,c^*}$ (main paper, Section 5) to studies that specify multiple competing predicted hypotheses.

S1 Relation between the Bayes factors against the unconstrained hypothesis and the complete complement

S1.1 Notation and preliminaries

For study t, let $H_1^{(t)}$ be the predicted hypothesis, $H_c^{(t)}$ its complement and $H_u^{(t)}$ the unconstrained hypothesis, and let $c_1^{(t)} \in (0, 1)$ be the complexity and $f_1^{(t)}$ the fit of $H_1^{(t)}$. As in Eqs. (4) and (5) of the main paper,

$$\mathrm{BF}^{(t)}_{1,u} = \frac{f_1^{(t)}}{c_1^{(t)}}, \qquad \mathrm{BF}^{(t)}_{c,u} = \frac{1 - f_1^{(t)}}{1 - c_1^{(t)}}, \qquad \mathrm{BF}^{(t)}_{1,c} = \frac{\mathrm{BF}^{(t)}_{1,u}}{\mathrm{BF}^{(t)}_{c,u}}. \tag{S1}$$

As in Appendix A of the main paper, the joint complexity and fit are $c_1^{(\text{joint})} = \prod_t c_1^{(t)}$ and $f_1^{(\text{joint})} = \prod_t f_1^{(t)}$. Since $f_1^{(t)} = c_1^{(t)} \mathrm{BF}^{(t)}_{1,u}$ by Eq. (S1), $f_1^{(\text{joint})} = c_1^{(\text{joint})}\, \mathrm{BF}^{(\text{joint})}_{1,u}$. Moreover, $H_u^{(t)}$ is the mixture of $H_1^{(t)}$ and $H_c^{(t)}$ with weights $c_1^{(t)}$ and $1 - c_1^{(t)}$, so that Eq. (S1) implies

$$c_1^{(t)}\, \mathrm{BF}^{(t)}_{1,u} + \left(1 - c_1^{(t)}\right) \mathrm{BF}^{(t)}_{c,u} = 1. \tag{S2}$$

By transitivity (main paper, Eq. (12)), $\mathrm{BF}^{(\text{joint})}_{c^*,u} = \mathrm{BF}^{(\text{joint})}_{1,u} / \mathrm{BF}^{(\text{joint})}_{1,c^*}$ is the ratio of the two joint Bayes factors. We say that the two Bayes factors converge if $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ approaches one as studies are added, that is, if they agree in relative terms (though their absolute difference need not vanish).

S1.2 Complexity weighting

Under complexity weighting, the derivation of Eq. (14) in Appendix A of the main paper shows (in its second-to-last step) that $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ is the fit of the complete complement divided by its complexity,

$$\mathrm{BF}^{(\text{joint})}_{c^*,u} = \frac{1 - f_1^{(\text{joint})}}{1 - c_1^{(\text{joint})}}. \tag{S3}$$

Direction. By Eq. (S3), $\mathrm{BF}^{(\text{joint})}_{c^*,u} < 1$ if and only if $f_1^{(\text{joint})} > c_1^{(\text{joint})}$, that is, if and only if $\mathrm{BF}^{(\text{joint})}_{1,u} > 1$. Hence, $\mathrm{BF}^{(\text{joint})}_{1,c^*} = \mathrm{BF}^{(\text{joint})}_{1,u} / \mathrm{BF}^{(\text{joint})}_{c^*,u}$ is larger than $\mathrm{BF}^{(\text{joint})}_{1,u}$ if $\mathrm{BF}^{(\text{joint})}_{1,u} > 1$ and smaller if $\mathrm{BF}^{(\text{joint})}_{1,u} < 1$, so that the two Bayes factors always point in the same direction and $\mathrm{BF}^{(\text{joint})}_{1,c^*}$ is at least as extreme as $\mathrm{BF}^{(\text{joint})}_{1,u}$.

Conditions for $\mathrm{BF}^{(\text{joint})}_{c^*,u} \approx 1$. The joint complexity $c_1^{(\text{joint})}$ shrinks geometrically as studies are added (if the complexities are bounded away from one), so the denominator tends to one and $\mathrm{BF}^{(\text{joint})}_{c^*,u} \approx 1 - f_1^{(\text{joint})}$. Hence, once $c_1^{(\text{joint})}$ is small, $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ is close to one if and only if the joint fit $f_1^{(\text{joint})}$ is small, that is, if it is unlikely that all predictions hold. The joint fit is a product of probabilities and decreases with every added study, but only slowly if the data make the predictions nearly certain in every study. In that case, the posterior concentrates in the joint predicted hypothesis, and $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ remains clearly below one.

S1.3 Equal weighting

Derivation. Under equal weighting, the expression for $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ in Appendix A of the main paper can be rewritten as

$$\begin{aligned}
\mathrm{BF}^{(\text{joint})}_{c^*,u} &= \frac{\prod_t \left(\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}\right) - \prod_t \mathrm{BF}^{(t)}_{1,u}}{2^T - 1} && \text{main paper, Eq. (16)} \\
&= \frac{\prod_t \left(\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}\right)}{2^T - 1} \left[1 - \prod_t \frac{\mathrm{BF}^{(t)}_{1,u}}{\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}}\right] && \text{factor out } \textstyle\prod_t \left(\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}\right) \\
&= \prod_t \frac{\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}}{2} \cdot \frac{1}{1 - 2^{-T}} \left[1 - \prod_t \frac{\mathrm{BF}^{(t)}_{1,u}}{\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}}\right] && \text{write } 2^T - 1 = 2^T\left(1 - 2^{-T}\right) \\
&= \prod_{t=1}^{T} \frac{\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}}{2} \cdot \frac{1 - \prod_{t=1}^{T} \tilde{f}^{(t)}}{1 - 2^{-T}} && \text{substitute } \tilde{f}^{(t)}, \qquad \text{(S4)}
\end{aligned}$$

where $\tilde{f}^{(t)} = \mathrm{BF}^{(t)}_{1,u} / \left(\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}\right) = \mathrm{BF}^{(t)}_{1,c} / \left(1 + \mathrm{BF}^{(t)}_{1,c}\right)$. Both factors can be interpreted in terms of how the 2^T components H_Ω of the joint parameter space (main paper, Section 5.2), including $H_1^{(\text{joint})}$ (i.e., $\Omega = \emptyset$), are weighted. Expanding the product as in Appendix A of the main paper shows that the first factor, $\prod_t \left(\mathrm{BF}^{(t)}_{1,u} + \mathrm{BF}^{(t)}_{c,u}\right)/2$, is the average of the Bayes factors $\mathrm{BF}^{(\text{joint})}_{\Omega,u}$ of all 2^T components (main paper, Eq. (6)), each with weight 2^{-T}. Weighting the same components by their complexities c_Ω instead gives $\prod_t \left[c_1^{(t)}\, \mathrm{BF}^{(t)}_{1,u} + \left(1 - c_1^{(t)}\right) \mathrm{BF}^{(t)}_{c,u}\right] = 1$ by Eq. (S2), which is the Bayes factor of $H_u^{(\text{joint})}$ against itself. The first factor thus compares equal with complexity weighting across the full joint parameter space. It reflects that the complete complement under equal weighting and the joint unconstrained hypothesis weight the components differently,

even though they cover nearly the same region of the joint parameter space. The second factor accounts for removing $H_1^{(\text{joint})}$ from the equally weighted average, where $\prod_t \tilde{f}^{(t)}$ is the share of $H_1^{(\text{joint})}$ in this average and 2^{-T} is its weight. Eq. (S3) has the same form under complexity weighting, where the share of $H_1^{(\text{joint})}$ in the complexity-weighted average is $f_1^{(\text{joint})}$ and its weight is $c_1^{(\text{joint})}$.

Conditions for $\mathbf{BF}_{c^*,u}^{(\text{joint})} \approx 1$. By Eq. (S4), this holds if both factors are close to one. Deviations in opposite directions, within or between the factors, can offset each other, but this requires a specific combination of complexities and evidence and is not a systematic tendency.

- *First factor.* The deviation of the term of study t from one is

$$\begin{aligned}
\frac{\mathrm{BF}_{1,u}^{(t)} + \mathrm{BF}_{c,u}^{(t)}}{2} - 1 &= \frac{\mathrm{BF}_{1,u}^{(t)} + \mathrm{BF}_{c,u}^{(t)}}{2} - \left[c_1^{(t)}\,\mathrm{BF}_{1,u}^{(t)} + \left(1 - c_1^{(t)}\right)\mathrm{BF}_{c,u}^{(t)}\right] && \text{Eq. (S2)} \\
&= \left(\tfrac{1}{2} - c_1^{(t)}\right)\mathrm{BF}_{1,u}^{(t)} - \left(\tfrac{1}{2} - c_1^{(t)}\right)\mathrm{BF}_{c,u}^{(t)} && \text{collect terms} \\
&= \left(\tfrac{1}{2} - c_1^{(t)}\right)\left(\mathrm{BF}_{1,u}^{(t)} - \mathrm{BF}_{c,u}^{(t)}\right) && \text{factor out } \tfrac{1}{2} - c_1^{(t)}. \quad \text{(S5)}
\end{aligned}$$

  Since $\mathrm{BF}_{1,u}^{(t)} > \mathrm{BF}_{c,u}^{(t)}$ if and only if $\mathrm{BF}_{1,c}^{(t)} > 1$, the term equals one if and only if $c_1^{(t)} = \frac{1}{2}$ or $\mathrm{BF}_{1,c}^{(t)} = 1$ (or both). It is above one if a narrow hypothesis is supported ($c_1^{(t)} < \frac{1}{2}$, $\mathrm{BF}_{1,c}^{(t)} > 1$) or a wide one is contradicted ($c_1^{(t)} > \frac{1}{2}$, $\mathrm{BF}_{1,c}^{(t)} < 1$), and below one in the opposite cases. Hence, the first factor is close to one if in each study the complexity is close to $\frac{1}{2}$ or the evidence is weak (or both). Because the terms multiply, however, even small deviations accumulate: If the terms are consistently above one, the first factor grows with every added study, and if they are consistently below one, it shrinks with every added study.

- *Second factor.* The second factor depends only on $\mathrm{BF}_{1,c^*}^{(\text{joint})}$ and T:

$$\begin{aligned}
\frac{1 - \prod_t \tilde{f}^{(t)}}{1 - 2^{-T}} &= \frac{1 - 1\big/\prod_t\big(1 + 1/\mathrm{BF}_{1,c}^{(t)}\big)}{1 - 2^{-T}} && \text{substitute } \tilde{f}^{(t)} = 1\big/\big(1 + 1/\mathrm{BF}_{1,c}^{(t)}\big) \\
&= \frac{1 - \mathrm{BF}_{1,c^*}^{(\text{joint})}\big/\big(\mathrm{BF}_{1,c^*}^{(\text{joint})} + 2^T - 1\big)}{1 - 2^{-T}} && \text{main paper, Eq. (11), solved for the product} \\
&= \frac{\big(2^T - 1\big)\big/\big(\mathrm{BF}_{1,c^*}^{(\text{joint})} + 2^T - 1\big)}{\big(2^T - 1\big)/2^T} && \text{simplify numerator and denominator} \\
&= \frac{2^T}{2^T - 1 + \mathrm{BF}_{1,c^*}^{(\text{joint})}} && \text{cancel } 2^T - 1. \quad \text{(S6)}
\end{aligned}$$

  Once there are more than a handful of studies, it is close to one if and only if $\mathrm{BF}_{1,c^*}^{(\text{joint})}$ is small relative to $2^T - 1$, the number of components of the complete complement. For example, for five studies with $\mathrm{BF}_{1,c}^{(t)} = 100$, $\mathrm{BF}_{1,c^*}^{(\text{joint})} \approx 608$ and the second factor is $32/639 \approx 0.05$. In contrast, for 20 studies with $\mathrm{BF}_{1,c}^{(t)} = 10$, $\mathrm{BF}_{1,c^*}^{(\text{joint})} \approx 1.8 \times 10^5$ is much larger, but the second factor is still 0.85, because $2^{20} \approx 10^6$.

- *Special case.* If $c_1^{(t)} = \frac{1}{2}$ in all studies, the equal weights coincide with the complexity weights (main paper, Section 5.3). The first factor then equals one by Eq. (S5), $\tilde{f}^{(t)} = f_1^{(t)}$ and $2^{-T} = c_1^{(\text{joint})}$, so that

Eq. (S4) reduces to Eq. (S3) and the results in Section S1.2 apply.

Consequences. Under equal weighting, the two properties derived for complexity weighting do not hold in general. First, $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ need not approach one, so that the two Bayes factors can differ by orders of magnitude even when many studies are synthesized. For example, if every study has $c_1^{(t)} = 0.10$ and $\mathrm{BF}^{(t)}_{1,c} = 10$, all terms of the first factor are above one, and $\mathrm{BF}^{(\text{joint})}_{c^*,u} = \mathrm{BF}^{(\text{joint})}_{1,u} / \mathrm{BF}^{(\text{joint})}_{1,c^*}$ equals 80 for $T = 5$, 2.5×10^4 for $T = 10$, and 1.5×10^9 for $T = 20$. This also shows that $\mathrm{BF}^{(\text{joint})}_{1,c^*}$ need not be at least as extreme as $\mathrm{BF}^{(\text{joint})}_{1,u}$ (e.g., $\mathrm{BF}^{(\text{joint})}_{1,c^*} = 642$ but $\mathrm{BF}^{(\text{joint})}_{1,u} = 1.6 \times 10^7$ for $T = 10$). If every study instead has $c_1^{(t)} = 0.90$ and $\mathrm{BF}^{(t)}_{1,c} = 10$, the ratio deviates in the opposite direction, with values of 0.03, 0.004, and 3.6×10^{-5}. Second, the two Bayes factors can point in opposite directions. For two studies, one with $c_1^{(1)} = 0.01$ and $\mathrm{BF}^{(1)}_{1,c} = 1000$ and the other with $c_1^{(2)} = 0.50$ and $\mathrm{BF}^{(2)}_{1,c} = 0.02$, $\mathrm{BF}^{(\text{joint})}_{1,u} = 3.57$, whereas $\mathrm{BF}^{(\text{joint})}_{1,c^*} = 0.06$ under equal weighting (and 3.61 under complexity weighting). In these examples, $\mathrm{BF}^{(t)}_{1,u} = \mathrm{BF}^{(t)}_{1,c} / \left(c_1^{(t)} \mathrm{BF}^{(t)}_{1,c} + 1 - c_1^{(t)}\right)$ by Eqs. (S1) and (S2).

S2 Multiple predicted hypotheses per study

S2.1 Predicted and alternative hypotheses

We assume that all studies test the same M competing theoretical accounts, and that the predicted hypotheses within a study are non-overlapping (overlapping hypotheses, as in the right panel of Figure S1, may be split into non-overlapping parts first). The constraints of the predicted hypotheses $H_m^{(t)}$ may differ across studies (as in Example B of the main paper), but each index m must refer to the same account in all studies, so that each joint hypothesis (e.g., that H_1 holds in all studies) represents a substantive theoretical claim. In the following example, all studies share the exact same predicted hypotheses, that is, the same constraints on the same parameters, and use the same prior. A linear mixed model with a fixed mean slope β and a covariance $\sigma_{\alpha,\beta}$ between random intercepts and slopes could, for instance, yield the competing predicted hypotheses[1]

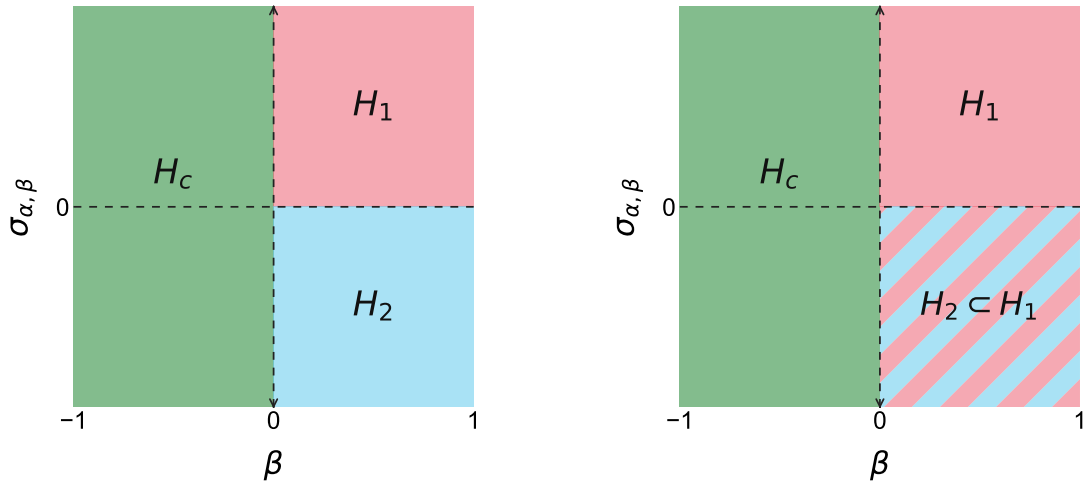
$$
\begin{aligned}
H_1^{(t)} &: \beta^{(t)} > 0, \quad \sigma_{\alpha,\beta}^{(t)} > 0 \\
H_2^{(t)} &: \beta^{(t)} > 0, \quad \sigma_{\alpha,\beta}^{(t)} < 0 \\
H_3^{(t)} &: \beta^{(t)} > 0, \quad \sigma_{\alpha,\beta}^{(t)} = 0.
\end{aligned}
$$

We denote the complexity and fit of $H_m^{(t)}$ by $c_m^{(t)}$ and $f_m^{(t)}$. Because $H_3^{(t)}$ contains an equality constraint, it has zero complexity, so that the encompassing Bayes factor $f_3^{(t)}/c_3^{(t)}$ is undefined (main paper, Section 5) and the computations below do not apply to $H_3^{(t)}$. Therefore, in the following, we consider only $H_1^{(t)}$ and $H_2^{(t)}$ (i.e., $M = 2$). However, since $H_3^{(t)}$ occupies no volume in the full parameter space, this does not affect the complement defined below.

The unconstrained hypothesis $H_u^{(t)}$ leaves both parameters unrestricted within their admissible ranges. In turn, the complement hypothesis comprises all parameter values not covered by *any* predicted hypothesis, that

[1] These three hypotheses are adapted from Zondervan-Zwijnenburg et al. (2020), who specified six predicted hypotheses. We use a subset of three and rename the parameters.

Figure S1: Illustration of multiple distinct (left panel) and overlapping (right panel) predicted hypotheses within a single study.



Note. The left panel shows the predicted hypotheses H_1 to H_3 with the complement alternative H_c. H_3 is not visible since it contains an equality constraint and is thus of lower dimensionality, covering zero area in the study's full parameter space. In the right panel, the hypotheses H_2 and H_c are the same, but H_1 now covers a larger part of the parameter space and includes H_2.

is, the complement of their union (left panel of Figure S1):

$$H_c^{(t)} : \beta^{(t)} < 0, \quad |\sigma_{\alpha,\beta}^{(t)}| < \sigma_\alpha^{(t)} \cdot \sigma_\beta^{(t)}.$$

Here, $\sigma_\alpha^{(t)}$ and $\sigma_\beta^{(t)}$ are the standard deviations of the random intercepts and slopes, so that the second condition merely states that the covariance can take on any value within its admissible range. Hence, $H_c^{(t)}$ differs from $H_u^{(t)}$ only in the constraint on the slope.

Together, $H_1^{(t)}, \ldots, H_M^{(t)}$ and $H_c^{(t)}$ partition the parameter space of each study into $M + 1$ regions, so that the joint parameter space consists of $(M + 1)^T$ combinations of regions across studies. The Bayes factor of each combination against the joint unconstrained hypothesis is the product of the study-specific Bayes factors of the regions involved (main paper, Eq. (6)). As in the main paper, the Bayes factor against the complete complement is a weighted average of the Bayes factors of the combinations it comprises (main paper, Eq. (7)), where the weights are the prior probabilities of these combinations under the assumption that the complete complement holds. We use equal weighting, which gives the same weight to each combination within a complete complement. This implies that evidence against the theoretical predictions is equally detrimental to the theory, regardless of which study provides it (main paper, Section 5.4).

S2.2 Testing a single joint predicted hypothesis against its own complete complement

Multiplying the study-specific Bayes factors yields one joint Bayes factor per predicted hypothesis, which can be interpreted as described in Section 4 of the main paper. To test a single $H_m^{(\text{joint})}$ against its own complete complement $H_{\overline{m}^*}^{(\text{joint})}$, which states that $H_m^{(t)}$ is false in at least one study, we proceed as in Sections 5.1 to 5.3

of the main paper. Within study t, the complement $H_{\overline{m}}^{(t)}$ of $H_m^{(t)}$ now comprises all other predicted hypotheses together with $H_c^{(t)}$; in our example, $H_{\overline{1}}^{(t)}$ comprises $H_2^{(t)}$ and $H_c^{(t)}$ (up to sets of measure zero). The components of $H_{\overline{m}^*}^{(\text{joint})}$ are all $(M+1)^T - 1$ combinations of regions other than $H_m^{(\text{joint})}$, including the other joint predicted hypotheses $H_k^{(\text{joint})}$ ($k \neq m$). Giving each component the weight $1/\big((M+1)^T - 1\big)$ and proceeding as in Appendix A of the main paper, $\text{BF}_{\overline{m}^*,u}^{(\text{joint})}$ follows from expanding $\prod_t \big(\sum_k \text{BF}_{k,u}^{(t)} + \text{BF}_{c,u}^{(t)}\big)$, which yields one term per combination, and subtracting the term $\prod_t \text{BF}_{m,u}^{(t)}$ of $H_m^{(\text{joint})}$ (for $\text{BF}_{c,u}^{(t)}$, see Eq. (S9)). By transitivity,[2]

$$\begin{aligned}\text{BF}_{m,\overline{m}^*}^{(\text{joint})} &= \frac{\big((M+1)^T - 1\big)\prod_{t=1}^T \text{BF}_{m,u}^{(t)}}{\prod_{t=1}^T \Big(\sum_{k=1}^M \text{BF}_{k,u}^{(t)} + \text{BF}_{c,u}^{(t)}\Big) - \prod_{t=1}^T \text{BF}_{m,u}^{(t)}} \\ &= \frac{(M+1)^T - 1}{\prod_{t=1}^T \Big(1 + \big(\sum_{k\neq m} \text{BF}_{k,u}^{(t)} + \text{BF}_{c,u}^{(t)}\big)/\text{BF}_{m,u}^{(t)}\Big) - 1}.\end{aligned} \tag{S7}$$

For $M = 1$, this reduces to Eq. (11) of the main paper.

S2.3 Comparing all joint predicted hypotheses simultaneously

To compare all joint predicted hypotheses simultaneously, we use posterior model probabilities (PMPs). In order for each PMP to reflect the probability that the respective hypothesis is true, the competing hypotheses must be mutually exclusive and exhaustive. The joint predicted hypotheses alone are not exhaustive. Adding the complete complement $H_{\overline{m}^*}^{(\text{joint})}$ of any one of them (Section S2.2) would make the set exhaustive but not mutually exclusive, because this complement contains the other joint predicted hypotheses. For example, for $M = 2$, the complete complement of $H_1^{(\text{joint})}$, which states that H_1 is false in at least one study, includes $H_2^{(\text{joint})}$, which states that H_2 holds in all studies. To obtain a set of hypotheses that is both mutually exclusive and exhaustive, we thus add the complete complement $H_{c^*}^{(\text{joint})}$, defined as the complement of the union of all joint predicted hypotheses $H_1^{(\text{joint})}, \ldots, H_M^{(\text{joint})}$. For $M = 1$, this is the complete complement as described in the main paper (Section 4). For $M > 1$, it comprises all parameter combinations in which no single H_m holds across all studies, for example, combinations in which H_1 holds in some studies and H_2 in the others, or in which H_c holds in at least one study. More precisely, of the $(M+1)^T$ combinations of regions, the M combinations in which the same $H_m^{(t)}$ holds in all studies are the joint predicted hypotheses, and the remaining $(M+1)^T - M$ combinations constitute $H_{c^*}^{(\text{joint})}$. As in the main paper, $H_{c^*}^{(\text{joint})}$ is not the conjunction of the study-specific complements $H_c^{(t)}$, which constitutes only one of its combinations. Since the predicted hypotheses within a study do not overlap (Section S2.1), the joint predicted hypotheses are mutually exclusive too. To compute PMPs, we require Bayes factors of all $M+1$ hypotheses $H_1^{(\text{joint})}, \ldots, H_M^{(\text{joint})}$ and $H_{c^*}^{(\text{joint})}$ against a common reference hypothesis. Two references are convenient: $H_u^{(\text{joint})}$ and $H_{c^*}^{(\text{joint})}$ itself.

With $H_u^{(\text{joint})}$ as reference, the Bayes factors for the joint predicted hypotheses can be computed as the products $\text{BF}_{m,u}^{(\text{joint})} = \prod_t \text{BF}_{m,u}^{(t)}$. As in the main paper, the Bayes factor for the complete complement is a

[2] A shortcut would be to apply the steps of the main paper to $H_m^{(t)}$ and $H_{\overline{m}}^{(t)}$ directly, that is, to compute $\text{BF}_{\overline{m},u}^{(t)} = \big(1 - f_m^{(t)}\big)/\big(1 - c_m^{(t)}\big)$ and to insert $\text{BF}_{m,\overline{m}}^{(t)} = \text{BF}_{m,u}^{(t)}/\text{BF}_{\overline{m},u}^{(t)}$ into Eq. (11) of the main paper. However, this shortcut does not give the same weight to every combination of regions across studies, as Eq. (S7) does, but only to each pattern of studies in which H_m fails. Within a study in which H_m fails, it splits this weight across the other predicted hypotheses and $H_c^{(t)}$ according to their complexities.

weighted average of the Bayes factors of its combinations, except that the complete complement now excludes all M joint predicted hypotheses rather than one. Under equal weighting, each of its $(M+1)^T - M$ combinations receives the weight $1/\big((M+1)^T - M\big)$, so that $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ is the average of their Bayes factors. Expanding the product $\prod_t \big(\sum_m \mathrm{BF}^{(t)}_{m,u} + \mathrm{BF}^{(t)}_{c,u}\big)$ yields one term for each of the $(M+1)^T$ combinations, so that subtracting the M terms of the joint predicted hypotheses gives

$$\mathrm{BF}^{(\text{joint})}_{c^*,u} = \frac{\prod_{t=1}^{T}\Big(\sum_{m=1}^{M} \mathrm{BF}^{(t)}_{m,u} + \mathrm{BF}^{(t)}_{c,u}\Big) - \sum_{m=1}^{M}\prod_{t=1}^{T} \mathrm{BF}^{(t)}_{m,u}}{(M+1)^T - M}. \tag{S8}$$

For $M = 1$, this reduces to Eq. (16) of the main paper. Eq. (S8) requires both $\mathrm{BF}^{(t)}_{m,u}$ of the predicted hypotheses and $\mathrm{BF}^{(t)}_{c,u}$ of the complement of their union from each study. If $\mathrm{BF}^{(t)}_{c,u}$ is not available from software, it can be computed from the complexities and $\mathrm{BF}^{(t)}_{m,u}$ of the predicted hypotheses, by generalizing Eq. (S1):

$$\mathrm{BF}^{(t)}_{c,u} = \frac{1 - \sum_{m=1}^{M} c^{(t)}_m \,\mathrm{BF}^{(t)}_{m,u}}{1 - \sum_{m=1}^{M} c^{(t)}_m}. \tag{S9}$$

With $H^{(\text{joint})}_{c^*}$ as the reference, the Bayes factors of the joint predicted hypotheses are $\mathrm{BF}^{(\text{joint})}_{m,c^*} = \mathrm{BF}^{(\text{joint})}_{m,u}/\mathrm{BF}^{(\text{joint})}_{c^*,u}$. Alternatively, $\mathrm{BF}^{(\text{joint})}_{m,c^*}$ can be written in terms of the study-specific Bayes factors $\mathrm{BF}^{(t)}_{k,c}$ of each predicted hypothesis $k = 1, \ldots, M$ against the complement of the union:

$$\mathrm{BF}^{(\text{joint})}_{m,c^*} = \frac{\big((M+1)^T - M\big)\prod_{t=1}^{T} \mathrm{BF}^{(t)}_{m,c}}{\prod_{t=1}^{T}\Big(1 + \sum_{k=1}^{M} \mathrm{BF}^{(t)}_{k,c}\Big) - \sum_{k=1}^{M}\prod_{t=1}^{T} \mathrm{BF}^{(t)}_{k,c}}. \tag{S10}$$

Eq. (S10) follows from dividing both $\mathrm{BF}^{(\text{joint})}_{m,u} = \prod_t \mathrm{BF}^{(t)}_{m,u}$ and $\mathrm{BF}^{(\text{joint})}_{c^*,u}$ in Eq. (S8) by $\prod_t \mathrm{BF}^{(t)}_{c,u}$, which leaves their ratio $\mathrm{BF}^{(\text{joint})}_{m,c^*}$ unchanged, and using $\mathrm{BF}^{(t)}_{k,c} = \mathrm{BF}^{(t)}_{k,u}/\mathrm{BF}^{(t)}_{c,u}$. For $M = 1$, Eq. (S10) reduces to Eq. (11) of the main paper. If the $\mathrm{BF}^{(t)}_{k,c}$ are available directly, this route requires neither $\mathrm{BF}^{(t)}_{c,u}$ nor the complexities.

With either reference $r \in \{u, c^*\}$, the PMP of each joint predicted hypothesis is

$$\mathrm{PMP}_m = P\left(H^{(\text{joint})}_m \mid D\right) = \frac{P\left(H^{(\text{joint})}_m\right)\,\mathrm{BF}^{(\text{joint})}_{m,r}}{\sum_{k=1}^{M} P\left(H^{(\text{joint})}_k\right)\,\mathrm{BF}^{(\text{joint})}_{k,r} + P\left(H^{(\text{joint})}_{c^*}\right)\,\mathrm{BF}^{(\text{joint})}_{c^*,r}}, \tag{S11}$$

where $\mathrm{BF}^{(\text{joint})}_{c^*,c^*} = 1$ if $r = c^*$, $P\big(H^{(\text{joint})}_m\big)$ and $P\big(H^{(\text{joint})}_{c^*}\big)$ are the prior probabilities of the joint hypotheses, and $\mathrm{PMP}_{c^*} = 1 - \sum_m \mathrm{PMP}_m$. One possibility is to assign $1/(M+1)$ to each of the $M+1$ joint hypotheses, which gives $H^{(\text{joint})}_{c^*}$ the same prior probability as each joint predicted hypothesis. Note that the prior probabilities of the joint hypotheses only enter the PMPs, whereas the weights of the components of $H^{(\text{joint})}_{c^*}$ enter the Bayes factors involving $H^{(\text{joint})}_{c^*}$.

References

Zondervan-Zwijnenburg, M., Richards, J., Kevenaar, S., Becht, A., Hoijtink, H., Oldehinkel, A., Branje, S., Meeus, W., & Boomsma, D. (2020). Robust longitudinal multi-cohort results: The development of self-control during adolescence. *Developmental Cognitive Neuroscience*, *45*, 100817. https://doi.org/10.1016/j.dcn.2020.100817